Ethan W. Todd

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Education

University of Michigan — **Ann Arbor, MI**
Candidate, Ph.D. Physics — *2023 – Present (expected 2026)*
Fall — Fall
Advisors: Prof. Igor Jovanovic, Prof. James Wells
2025 Winter M.S. Physics

Florida State University — **Tallahassee, FL**
B.S. Physics, Summa Cum Laude — *2020 – 2023*
Fall — Spring
Honors Thesis: Study of Displaced Vertex Tagging with the CMS Experiment
Advisor: Prof. Ted Kolberg

Emory University — **Atlanta, GA**
Foundational Coursework — *2018 – 2020*
Fall — Spring

Publications

7. A. R. L. Kavner, **E. Todd**, M. Babicz, et al., "Super-Lindhard Ionization from 254 eV_{nr} Nuclear Recoils in Germanium," *under review*, 2026. [Online]. Available: https://arxiv.org/abs/2609.20848
6. CMS HGCAL Collaboration, "Leakage Current of High-Fluence Neutron-Irradiated 8" Silicon Sensors for the CMS Endcap Calorimeter Upgrade," *under review*, 2026. [Online]. Available: https://arxiv.org/abs/2510.17280
5. V. Fondement, J. Bae, F. Rahman, et al., "Digital pulse processing for mixed-field ionizing radiation detection in rare event nuclear experiments," *Fusion Science and Technology*, 2026. DOI: 10.1080/15361055.2026.2732349
4. V. Fondement, **E. Todd**, and I. Jovanovic, "Updated ^{78m}Br $2^- \rightarrow 1^+$ Lifetime from Isomeric Decay Cascade in Active Target," *Nuclear Physics A*, 2026. DOI: 10.1016/j.nuclphysa.2026.123485
3. **E. Todd**, V. Fondement, and I. Jovanovic, "Machine Learning-Enabled Pulse Shape Discrimination in YAP:Ce," *Nuclear Instruments and Methods in Physics Research Section A*, 2026. DOI: 10.1016/j.nima.2026.171883
2. C. Graham, **E. Todd**, C. Wilson, et al., "Light Output-Dependent Integration Gating for Optimized Pulse Shape Discrimination," *Nuclear Instruments and Methods in Physics Research Section A*, 2026. DOI: 10.1016/j.nima.2026.171521
1. K. Yocum, H. Smith, **E. Todd**, et al., "Millimeter/Submillimeter Spectroscopic Detection of Desorbed Ices: A New Technique in Laboratory Astrochemistry," *The Journal of Physical Chemistry A*, 2019. DOI: 10.1021/acs.jpca.9b04587

Research Experience

Low Energy Physics Group, Argonne National Laboratory — **Lemont, IL**
Graduate Research Aide — *2026 – Present*
Summer

Measurement of the ^{37}Cl(α, n)^{40}K Cross Section

Supervisors: Dr. Daniel Santiago-Gonzalez, Dr. Melina Avila

- Performed experimental measurements of the ^{37}Cl(α, n)^{40}K reaction using the Multi-Sampling Ionization Chamber (MUSIC) at the ATLAS accelerator facility at Argonne National Laboratory.
- Implemented an event reconstruction algorithm for prior MUSIC data that performs offline time synchronization of four digitizers via analysis of the time structure of the beam, and uses CUDA-accelerated time sorting to greatly increase data-processing throughput.
- Developed a method to correct systematic deviations in trapezoidal-filtered-pulse heights originating from a mismatch between the filter pole-zero and preamp decay time.

Applied Nuclear Science Group, University of Michigan — **Ann Arbor, MI**
Graduate Student Researcher — *2024 – Present*
Summer

Pulse Shape Discrimination in YAP:Ce via Machine Learning

- Compared α and γ-ray discrimination capabilities of charge comparison/shape indicator to several machine learning methods: random forests, gradient boosting, XGBoost, and multilayer perceptrons.
- Quantifed the performance of each method using receiver operating characteristic curves, and demonstrated the substantial advantage of machine learning over traditional methods.
- Applied the XGBoost model to proof-of-concept mixed field radiation environments consisting of ^{241}Am and ^{137}Cs or ^{241}Am and ^{60}Co, and demonstrated that known spectral features of ^{241}Am, ^{137}Cs, and ^{60}Co are correctly classified.

Precision Measurements of ^{72}Ge(n, γ) at Thermal Energies

- Utilized the Ohio State Nuclear Reactor Laboratory thermal neutron beam facility to perform new measurements of ^{72}Ge(n, γ)^{73}Ge gamma-ray transition energies with a focus on the 68.75 keV final-state transition.
- Used a position-sensitive H3D M400 CdZnTe detector to measure low-energy gamma-ray interactions with event-by-event position sensitivity.
- Developed custom analysis code to select single-scatter events near the detector surface, reducing the Compton background and improving sensitivity in the low-energy region.
- Provided an independent measurement of the 68.75 keV gamma-ray energy in order to verify a separate UZH/PSI gamma-ray measurement and further support the validity of a recent high 254 eV$_{nr}$ germanium ionization-yield result.

CMS Group, Florida State University — **Tallahassee, FL**
Undergraduate Student Researcher — *2021 – 2023*
Fall *Summer*

Search for BSM Long-lived Particles

Advisor: Prof. Ted Kolberg

- Utilized deep neural network machine learning to search for beyond the Standard Model long-lived particles in data from the CMS Experiment.
- Trained ML models on Monte Carlo data with varying signal (mass, lifetime) and background (t, b physics) to verify their ability to identify displaced vertices independently of the particular physics.
- Investigated the most effective parameters of ML models and verified physics-based expectations.

Silicon Sensor Quality Control

Advisor: Prof. Rachel Yohay

- Conducted process quality control (PQC) testing for the CMS High Granularity Calorimeter (HGCAL) to monitor the quality and stability of the silicon sensor production process.
- Analyzed and presented PQC results to the CMS HGCAL sensor testing group.
- Helped create and implement a standardized testing procedure at FSU.

WW Astrochemistry Group, Emory University — **Atlanta, GA**
Undergraduate Student Researcher — *2018 – 2020*
Fall *Fall*

Advisor: Prof. Susanna Widicus Weaver

- Performed millimeter/submillimeter spectroscopic measurements of desorbed astrophysical ice analogs.
- Conducted proof-of-concept millimeter/submillimeter spectroscopic measurements of H_2O and D_2O binding energy, sublimation enthalpy, and sublimation entropy.

Teaching Experience

University of Michigan — **Ann Arbor, MI**
Lead Graduate Student Instructor — *2025 - 2026*
Fall *Winter*

- Oversaw 30 graduate student instructors teaching general physics laboratories with a total enrollment of 1800 undergraduate students.
- Edited and maintained the laboratory manuals and Jupyter notebook worksheets used in the course.
- Managed the course information and organization via Canvas.
- Arbitrated student and graduate student instructor concerns.

University of Michigan — **Ann Arbor, MI**
Graduate Student Instructor — *2023 – 2025* (*Fall* *Spring*)

- 2025 Spring Physics 251, Fundamental Physics for the Life Sciences II Lab
- 2024 Fall Physics 121, Physics for Architects Lab
- 2023 Fall – 2024 Winter Physics 241, General Physics II Lab

Florida State University — **Tallahassee, FL**
Physics Learning Assistant — *2022* *Spring*

PHY 2049C, General Physics B

Emory University — **Atlanta, GA**
Laboratory Teaching Assistant — *2019* *Fall*

CHEM 150L, Structure and Properties Lab

Honors and Awards

2026 Winter: Marcellus L. Wiedenbeck Teaching Award — *Awarded for exemplifying outstanding teaching abilities.*

2026 Winter: Rackham Outstanding Graduate Student Instructor Award — *Awarded for demonstrating extraordinary dedication and excellence as a teacher.*

2023 Spring: Phi Beta Kappa Society

2023 Spring: Joseph Lanutti Undergraduate Research Award — *Awarded for third place in the FSU physics departmental poster presentation contest.*

2022 Fall: Evelyn and John Baugh Research Presentation Scholarship — *Awarded to travel and present research at SESAPS.*

2022 Spring: Anna Runyan Undergraduate Endowment — *Awarded for outstanding academic success in physics.*

Grants

2025 Fall: NSUF Rapid Turnaround Experiment Grant — *Awarded to conduct precision measurements of Ge gamma-ray cascades at the Ohio State University Nuclear Reactor Laboratory.*

2025 Fall: Rackham Graduate Student Research Grant — *Awarded to conduct research at the Paul Scherrer Institute on the ionization yield of high-purity germanium detectors.*

Skills

Agentic AI: Local deployment of autonomous agents in opencode and son-of-anton, with open-weight models served via vllm, llama.cpp, and AMD-hardware-optimized engines (*e.g.,* unholy, gufo)

Programming: C++, Python, Nix, CUDA

Software:

Nuclear Measurements:	ROOT, Geant4, CoMPASS, wavedump, LISE^{++}, SRIM
Machine Learning:	Scikit-learn, XGBoost, TensorFlow
Version Control:	Git, Nix Flakes
Build Systems:	Nix, Make, CMake
Scientific Presentation:	LaTeX/Beamer

Hardware: CAEN digitizers, soldering, electronic testing equipment, *e.g.,* oscilloscopes

Presentations

5. **E. Todd**, V. Fondement, and I. Jovanovic, *Machine Learning-Enabled Pulse Shape Discrimination in Fast Inorganic Scintillators*, ANL PSE Student Lunch Talk Summer Series, Lemont, IL, 2026

4. **E. Todd**, V. Fondement, and I. Jovanovic, *Particle Identification in Fast Inorganic Scintillators via Machine Learning*, Fall Meeting of the Eastern Great Lakes Section of the American Physical Society, Ypsilanti, MI, 2025

3. **E. Todd** and I. Jovanovic, *Investigation of Anomalously High Ionization Produced by Low Energy Nuclear Recoils in Germanium (poster)*, DOE NNSA Office of Defense Nuclear Nonproliferation R&D University Program Review, Gainesville, FL, 2025

2. **E. Todd**, A. A. Kadhim, N. Bower, et al., *Process Quality Control for HGCAL at Florida State University (poster)*, 89th Annual Meeting of the Southeastern Section of the American Physical Society, Oxford, MS, 2022

1. **E. Todd**, K. Yocum, P. Gerakines, et al., *A Novel Use of Rotational Spectroscopy for Studying Characteristics of Desorbed Interstellar Ice Analogs (poster)*, Emory SURE 2019 Symposium, Atlanta, GA, 2019